

1. SPECIFICATION

1.1 Drawing No. : 330811-1

1.2 Mechanical data

1.2.1 Material Data : Refer to Material List Page 2

1.2.2 PCB Drawing : Refer to Page 3 – 4. 6

1.2.3 Dimensions : Refer to Mechanical Dimensions

1.2.4 Schematic Drawing : -

1.3 Electrical

Not Applicable

1.4 Visual Requirements

1.4.1 Refer to Incoming Inspection Procedure

1.5 Incoming Quality

1.5.1 Dimensional check

1.5.2 Visual Inspection

2. REFERENCE

2.1 Original P/N : 330811-1

2.2 Original Dwg No. : 326423 REV. E

: 330811 REV. B

2.3 Manufacture : SPINNAKER CONTRACT MFG INC. | 68489

2.4 Manufacture Part No. : -

3. DOCUMENT FROM SUPPLIER

3.1 Delivery Order

4. PACKING

Not Applicable

5. STORAGE / HANDLING

Not Applicable

REMARKS:

					NAME	DATE	DEPARTMENT	
				DRAWN BY	TRI SUBEKTI	30.11.2023	LIGHTING	
				CHECKED BY	PAUL SCHWAB	30.11.2023	ENGINEERING	
				DESCRIPTION				PART/ BAU NO
				330811-1/PC BOARD ASSEMBLY FOR LS-232				27400375
0	CMS37087	27.09.24	ARIEF					
REV	ECN NO	DATE	NAME	SUBSTITUTE FOR:		DATED:		PAGE 1 OF 6

MATERIAL LIST FOR PC BOARD ASSEMBLY FOR LS-232

No.	Original Part#	Material Description	Reference Designators	Qty	Approved Manufacturer List	
					Manufacturing	Manufacturing Part#
1	326423	PCB LS-232 LIGHT SYSTEM C326423	-	1	NULL	
2	CP-1279	CAPACITOR, 470uF, 35V, 20%, ALUMINUM	C1	1	NULL	
3	CP-1195	CAP 0.001uF 100V 5% POLYESTER METALIZED	C2, C4	2	NULL	
4	CP-1194	CAP 0.01uF 100V 5% POLYESTER METALIZED	C3	1	NULL	
5	CP-1062	CAPACITOR 1uF 50V 10% TANTALUM RADIAL	C5	1	TTI INC.	
6	CP-1228	CAP 0.1uF 200V 10% POLYESTER FILM	C6	1	NULL	
7	CP-1204	CAP 0.022uF 100V 5% POLYESTER METALIZED	C7	1	NULL	
8	AP-1341	DIODE SCHOTTKY BARRIER RECTIFIER	C4	1	NULL	
9	AP-1280	DIO MUR140 1A 400V ULTRAFAST AXIAL	CR1, CR2, CR3, CR7, CR8	5	NULL	
10	87-0003	DIO 1N4007 1000V 1A AXIAL	C9	1	NULL	
11	AP-1302	ZENER, 6.2V, 500mW, +/-5%, GLASS CASE	CR12	1	NULL	
12	87-0002	DIO HIGH CONDUCTANCE FASTDIODE	CR6, CR10	2	NULL	
13	AP-1331	DIO MICROPOWER VOLTAGE REFERENCE 1.2V 1%	CR11	1	NULL	
14	AP-1279	DIO GP60-30 3KV 6A FRONTIER ELECTRONICS	CR13	1	SHENZHEN KINGSUN ELECTRONICS CO.	122286
15	TP-1145	CHOKE 0.7uH	L2	1	NULL	
16	CP-1077	0.25uF 1200V 10% OIL-FILLED AX	C8	1	MOTOR CAPACITORS INC.	
17	AP-1327	XSTR MOSFET 200V 18A 0.18 OHM RDS ON	Q2	1	NULL	
18	AP-1252	SCR, SENSITIVE GATE, 400V, 4ARMS, CASE	Q3	1	NULL	
19	RP-1372	RES 1M 1/2W 1% MF AX	R1, R2	2	NULL	
20	RP-1308	RES 20.0K 1/4W 1% MF AX	R4	1	NULL	
21	RP-1178	RES 110K 1/4W 5% CF AX	R8	1	NULL	
22	RP-1202	RES 2.2K 1/4W 5% CF AX	R9	1	NULL	
23	RP-1253	RES 150 OHM 1/4W 5% CF	R5	1	NULL	
24	RP-1354	RES 3.3 OHM 1/4W 5% CF AX	R11	1	NULL	
25	RP-1360	RES 220 OHM 1/4W 5% CF AX	R19	1	NULL	
26	RP-1463	RES 0.10 OHM 1W 2% MO AX	R13	1	NULL	
27	RP-1376	RES 270 OHM 1/4W 5% CF AX	R12, R14	2	NULL	
28	RP-1366	RES 68 OHM 1/4W 5% CF AX	R15	1	NULL	
29	RP-1180	RES 470 OHM 1/4W 5% CF AX	R16	1	NULL	
30	RP-1320	RES 1K 1/2W 5% CF AX	R17	1	NULL	
31	RP-1363	RES 100 OHM 1/2W 5% CF AX	R18	1	NULL	
32	RP-1173	RES 3.3K 1/4W 5% CF	R3	1	NULL	
33	SP-1069	TOGGLE ON-OFF RIGHT-ANGLE PC THRU-HOLE	SW1, SW2	2	NULL	
34	TP-1142	XFMR CONVERTER, TR-2202, MFG SPEC.	T1	1	SHENZHEN KINGSUN ELECTRONICS CO.	122286
35	AP-1328	IC HYBRID CIRCUIT 20-PIN DIP (330671)	U1	1	SHENZHEN KINGSUN ELECTRONICS CO.	122286
36	AP-1309	IC OPTOCOUPLER PHOTOTRANSISTOR OUTPUT	U2	1	NULL	
37	AP-1332	IC5V,100mAVOLTAGEREGULATOR TO92PKG	U3	1	NULL	
38	EP-1355	CONNECTOR 9CONTACT D-SUB HIGH	J2	1	NULL	
39	EP-1277	HDR 3 PIN 0.10 SPACING STRAIGHT	J4	1	NULL	
40	EP-1278	HDR SHUNT 2-PIN 0.100 SPACING AU PLATED	XJ4	1	NULL	
41	EP-1361	CON SOCKET STRIP SSQ SERIES CENTERLINE:	H1	1	NULL	
42	EP-1362	CON SOCKET STRIP SSQ SERIES CENTERLINE:	H2	1	NULL	
43	EP-1363	DC INPUT POWER CONNECTOR	J3	1	NULL	
44	48-0170	SCREW 4-40 X 3/8 BH SS	XJ2	2	NULL	
45	48-2680	NUT 4-40 KEPS #511-041800-00 SHAKEPROOF	XJ2	2	NULL	
46	330815	FEED BACK WIRE FOR LS-232 PC BOARD	-	1	NULL	
47	330811	PC BOARD FOR LS-232 LIGHT SYSTEM	-	1		

		NAME	DATE	DEPARTMENT	
	DRAWN BY	TRI SUBEKTI	30.11.2023	LIGHTING	
	CHECKED BY	PAUL SCHWAB	30.11.2023	ENGINEERING	
	DESCRIPTION				PART/ BAU NO
	330811-1/PC BOARD ASSEMBLY FOR LS-232				27400375
SUBSTITUTE FOR:			DATED:		PAGE 2 OF 6

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ALL DIMENSION
TOLERANCE ARE
IN INCHES UNLESS
OTHERWISE SPECIFIED

MATERIAL	—
	—
FINISH	—
	—



THIRD ANGLE

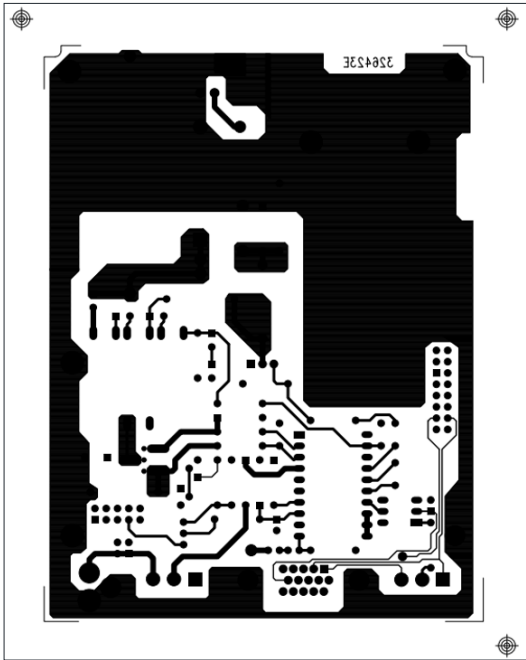
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ARE:		
FRACTIONS	DECIMALS	ANGLE
1/64	.XX ± .01	± 1°
	.XXX ± .005	
SURFACE ROUGHNESS:		

	NAME	DATE	DEPARTMENT			
DRAWN BY	TRI SUBEKTI	30.11.2023	LIGHTING			
CHECKED BY	PAUL SCHWAB	30.11.2023	ENGINEERING	PART/BAU NO		SIZE
DESCRIPTION 330811-1/PC BOARD ASSEMBLY FOR LS-232				27400375		A4
				WEIGHT		SCALE
						SHEET 3 OF 6

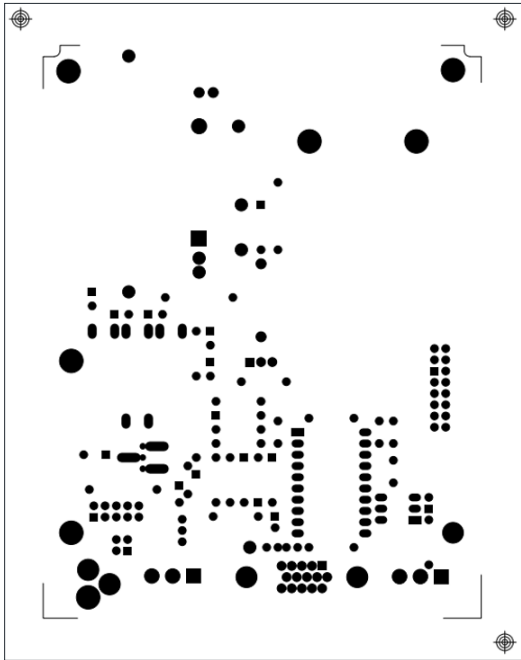
RAW MATERIAL SPECIFICATION

PCB LS-232 LIGHT SYSTEM (326423)

REV	ECN NO.	DESCRIPTION	NAME	DATE
0	CMS37087	INITIAL RELEASED	ARIEF	27.09.24



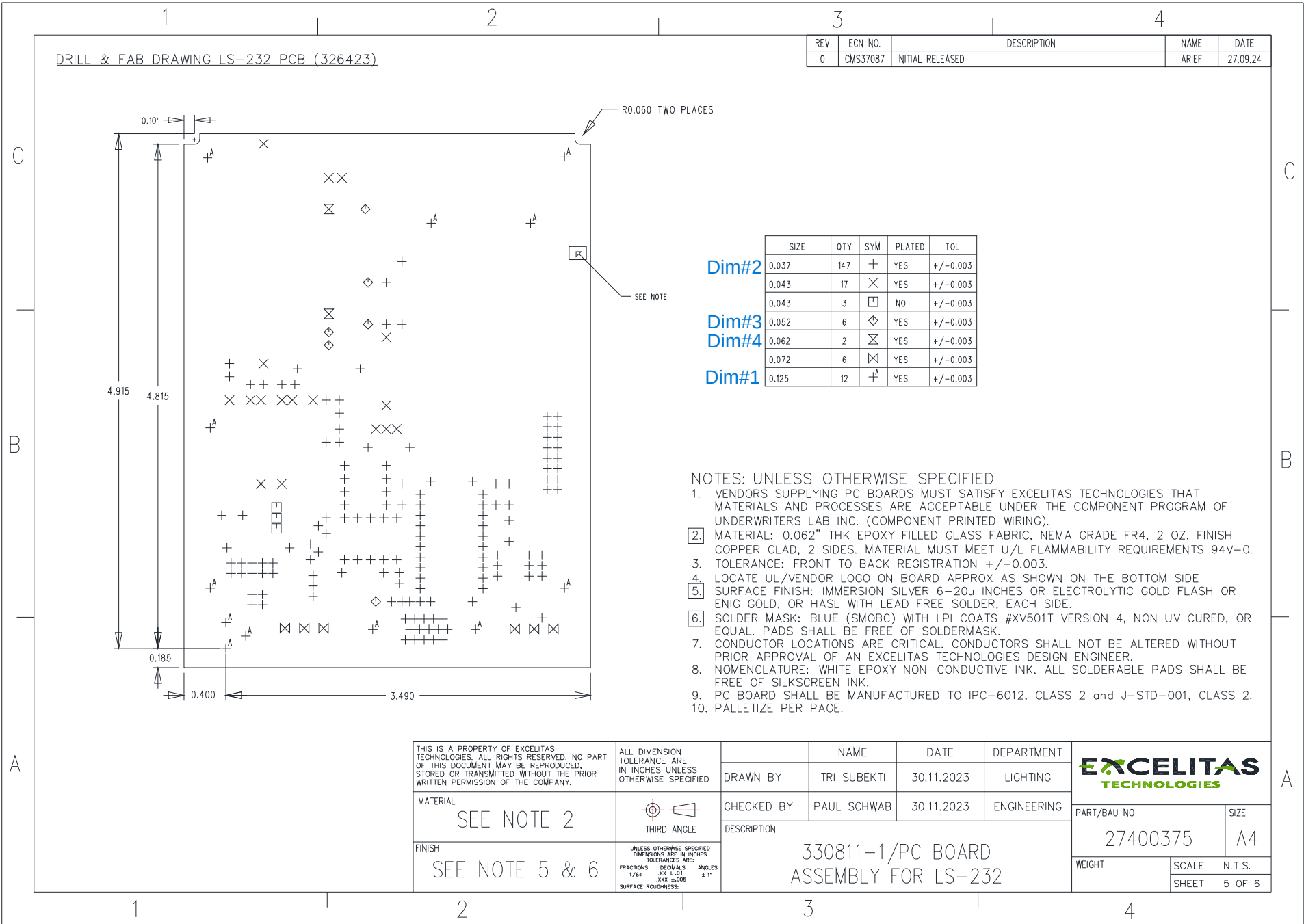
ARTWORK BOTTOM SIDE
LS-232 PCB



SOLDERMASK BOTTOM SIDE
LS-232 PCB

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		DRAWN BY	TRI SUBEKTI	30.11.2023	LIGHTING		
		CHECKED BY	PAUL SCHWAB	30.11.2023	ENGINEERING	PART/BAU NO	SIZE
MATERIAL		DESCRIPTION				27400375	A4
FINISH						WEIGHT	SCALE
		330811-1/PC BOARD ASSEMBLY FOR LS-232				SHEET	4 OF 6

RAW MATERIAL SPECIFICATION



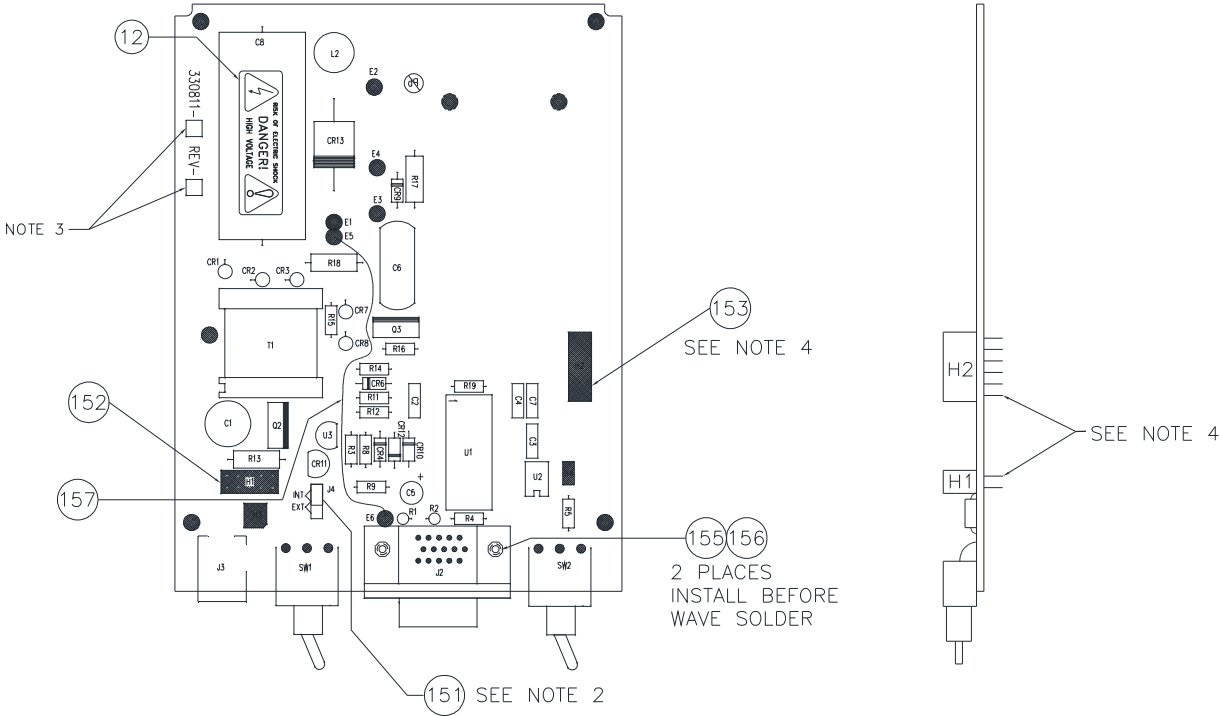
RAW MATERIAL SPECIFICATION

PC BOARD FOR LS-232 LIGHT SYSTEM (330811)

NOTES:

1. MASK E1-E6, J3, SW1 & SW2 AND FIVE (5) MOUNTING HOLES BEFORE WAVE SOLDER.
2. INSTALL ITEM #151 ON J4 IN INTERNAL POSITION.
3. MARK DASH NUMBER AND REVISION IN BLOCKS WITH INDELIBLE INK.
4. INSTALL ITEMS #152 AND #153, (H1 AND H2) AFTER WAVE SOLDER AND DO NOT CUT LEADS. (PINS ON SOLDER SIDE OF BOARD).
5. ASSEMBLY SHALL MEET IPC/E1A J-STD-001, CLASS 2 AND IPC-A-610, CLASS 2.

● -MASK BEFORE WAVE SOLDER.



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				DRAWN BY		TRI SUBEKTI		30.11.2023			
MATERIAL		 THIRD ANGLE		CHECKED BY		PAUL SCHWAB		30.11.2023		ENGINEERING	
FINISH				DESCRIPTION 330811-1/PC BOARD ASSEMBLY FOR LS-232							
		UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ARE: FRACTIONS DECIMALS ANGLES 1/64 .XX ± .01 ° ± 1° SURFACE ROUGHNESS:		PART/BAU NO 27400375 SIZE A4							